PCB-Loader — Standard Operating Procedure

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PCB-Loader — Standard Operating Procedure

> **Audience:** Maintenance Tech · Line Manager **Applies to:** Line-E, station 1, PCB-Loader

1. Purpose

The PCB-Loader is the entry point of Line-E (Electronics Assembly). It feeds bare PCBs from magazine racks onto the conveyor for SMT-Placement. A stoppage starves the entire 8-station line.

2. Normal operating envelope

- Ideal cycle time: **5 s** — fastest station in the factory
- Max acceptable cycle time: **7 s**
- Expected reject rate: **0 %**
- Fault probability per cycle: **0.1 %**
- Input buffer capacity: **5 parts**

3. Idle reasons

`idle_reason`	Meaning here	First check
`Starved`	Magazine racks empty.	Refill magazine; check Production Schedule.
`Blocked`	SMT-Placement (station 2) buffer full.	Verify SMT-Placement state.

4. Faults & corrective actions

4.1 `Feed-Jam`

- **Symptoms in telemetry:** `fault_type = "Feed-Jam"`. Loader stops; all downstream Line-E stations report `Starved`.
- **Likely root cause:** PCB stuck in feeder gate; magazine misaligned.
- **Corrective action:**
  1. LOTO. Clear the jammed PCB.

2. Verify magazine alignment.
- **Restart criteria:** 10 PCBs feed cleanly.

4.2 `Magazine-Empty`

- **Symptoms in telemetry:** `fault_type = "Magazine-Empty"`. Although it reads as a fault, this is essentially a refill condition.
- **Likely root cause:** kit consumed faster than expected.
- **Corrective action:**
  1. Refill the magazine.
  2. Confirm lot traceability captured.
- **Restart criteria:** continuous feed for 5 cycles.

4.3 `Sensor-Block`

- **Symptoms in telemetry:** `fault_type = "Sensor-Block"`. Pickoff sensor reads “present” when no PCB is at the gate.
- **Likely root cause:** sensor fouled, dust accumulation, or misalignment.
- **Corrective action:**
  1. Clean and realign the sensor.
  2. Verify trigger margin.
- **Restart criteria:** clean transitions on 10 cycles.

5. Preventive maintenance schedule

- **Daily:** check magazine; clean sensors.
- **Weekly:** lubricate feed mechanism.
- **Monthly:** alignment verification.
- **Quarterly:** mechanical overhaul.

6. References

- `Maintenance_Workflow_Overview.pdf`
- `Line-E_02_SMT-Placement_SOP.pdf`
- `Production_Lines_Reference.pdf`